

SUDHARSHAN MURUGAN

Data Engineer | Frontend Developer | AI/ML Enthusiast

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PROFESSIONAL SUMMARY

Data Engineer at Tata Consultancy Services with 1+ year of experience delivering enterprise data solutions for Liberty Mutual Insurance. M.Sc Computer Science graduate (CGPA 9.19) from SRM Institute and Gold Medalist in B.Sc CS (CGPA 9.55) from SCSVMV University. Skilled in building scalable ETL pipelines, data modeling, SQL optimization, and Python-based automation. Also experienced in frontend development and AI/ML model deployment, enabling end-to-end solution delivery across the technology stack.

PROFESSIONAL EXPERIENCE

Data Engineer

Tata Consultancy Services (TCS) — Liberty Mutual Insurance

Chennai, India • May 2025 – Present • 1+ Year

- Design and develop scalable ETL pipelines and data workflows for Liberty Mutual Insurance BI and analytics needs.
- Build and maintain data models, integrations, and transformations using SQL and Python in cloud environments.
- Collaborate with data analysts, architects, and business stakeholders to deliver enterprise data solutions.
- Monitor data quality, troubleshoot pipeline failures, and optimize performance for large-scale data processing.
- Contribute to documentation, data governance practices, and best-in-class engineering standards.

Front-End Developer Intern

Magic Bus India Foundation

Remote • 2023 – 2024

- Developed responsive web interfaces using HTML, CSS, and modern front-end frameworks.
- Reduced page load time by 30% through targeted performance optimizations.
- Created a dynamic homepage that boosted user engagement by 20%.
- Automated manual processes by integrating advanced web forms, saving significant staff hours.

Python & ML Developer Intern

Code Clause

Remote • 2023

- Designed and deployed Python-based tools including a password generator and calculator.
- Built and implemented AI models using Python for intelligent problem-solving with complex datasets.

KEY PROJECTS

Lung Cancer Detection System — PG Major Project

Python | Flask | TensorFlow | OpenCV | EfficientNet-B0 CNN

- Flask-based web app leveraging EfficientNet-B0 CNN to detect lung cancer from CT scans with high accuracy.
- Integrated data visualization for detailed prediction analysis; designed intuitive upload and analysis interface.
- Enabled healthcare professionals to perform preliminary screenings, reducing diagnostic time significantly.

Stock Management System — UG Mini Project

VB.NET | MySQL

- Developed a stock inventory management application with CRUD operations and real-time data updates.
- Integrated data visualization for inventory analysis and reporting, reducing manual errors.

EDUCATION

Master of Science in Computer Science

SRM Institute of Science and Technology, Kattankulathur, Chennai

CGPA: 9.19

2022 – 2024

TECHNICAL SKILLS

Data Engineering	ETL Pipeline Development, Data Modeling, SQL, Data Warehousing, Data Quality Management
Languages	Python, JavaScript, HTML, CSS, SQL, Java (basics)
Frameworks & Libraries	Flask, Bootstrap, TensorFlow, Scikit-learn, React.js (Learning)
Databases	MySQL, MongoDB (basics)
Tools & Platforms	Git, GitHub, VS Code, OpenCV, Cloud Platforms (working knowledge)
Web Development	Frontend Design, Responsive Web Development

CERTIFICATIONS & PARTICIPATION

- Python — Guvi Educational Company
- Supervised Machine Learning: Regression and Classification — Stanford Online / Coursera
- HTML, CSS & JavaScript Essential Training — LinkedIn Learning
- MySQL Basics — Great Learning Platform
- Core Java — Besant Technologies
- ISRO LULC Mapping Project using ML techniques — June 2023